

The last two weeks of FYRE I have learned a lot about microcontrollers. I first learned coding basics: variables, functions, and classes. I learned the different data types (mainly integers, strings, and booleans) and popular functions like `sleep()` or `print()`. I used that to learn how to control an Arduino ESP-32. I made programs in Micropython such as a blink program that makes the light from my Arduino flicker.

Now that I had a foundation in coding, I was able to give AI prompts for more sophisticated programs. AI, just like humans, make mistakes. Between learning how to use AI to ask for the correct prompts for programming and my basic understanding of micropython, I was able to prototype a security system and a raindrop sensor. I was also tasked with being able to create the correct circuit so the programming would work to create the raindrop sensor and the security system. I already from previous weeks had a basic understanding of how to build a circuit, but my understanding definitely is stronger now. I worked with pretty much every basic component of circuits: breadboards, resistors, premade sensors, potentiometers, LEDs.

The first project I worked on was the security system, and the challenge I had with it was figuring out how I could use the components I was given in a way that would create the prototype, me and my partners solved this by utilizing a tactic we had learned earlier in class. We drew a diagram of our system from start to finish. It helped us identify the things we needed. Using the breadboard we put in a button that when pressed it would turn on one of the LEDs. Our other switch was the sensor, it would turn on both of the LEDs. The button was used to “disable” the alarm and the sensor would “activate the alarm”. All components were connected to a pin on the arduino which allowed us to control our components.

The sensor project was a lot easier as we were given an image of the circuit. My partners and I did encounter some confusion about where things exactly went in the circuit as it was hard to pick up from the photo. My role for this project was mainly helping one of my partners to determine what direction and where things go on the circuit board. We were the ones responsible for creating the circuit that would detect the change in voltage when introduced to wet wipes or us breathing on it.